Computer Science **Lab C – QUICK SORT**

Lab Goal : This lab was designed to teach you more about sorting and searching and specifically about quick sort.

Lab Description : Write a program that demonstrates you know how to implement a quick sort.

Sample Data :

```
9 5 3 2
19 52 3 2 7 21
68 66 11 2 42 31
```

Sample Output :

```
pass 0 [2, 5, 3, 9]

pass 1 [2, 5, 3, 9]

pass 2 [2, 3, 5, 9]


pass 0 [7, 2, 3, 52, 19, 21]

pass 1 [3, 2, 7, 52, 19, 21]

pass 2 [2, 3, 7, 52, 19, 21]

pass 3 [2, 3, 7, 21, 19, 52]

pass 4 [2, 3, 7, 19, 21, 52]


pass 0 [31, 66, 11, 2, 42, 68]

pass 1 [2, 11, 66, 31, 42, 68]

pass 2 [2, 11, 66, 31, 42, 68]

pass 3 [2, 11, 42, 31, 66, 68]

pass 4 [2, 11, 31, 42, 66, 68]
```

quickSort Algorithm

```
method quicksort(array,start,end)
   as long as start is less than end
      split = partition(array, start, end)
      quicksort(array, start, split-1)
      quicksort(array, split+1, end)


method partition(array,start,end)
   pivot is the left most value in array
   left = start
   right = end

   while left is less than right
      Increment left until at right or a large item is found
      Increment right until past left or a small/equal item is found
      if left < right
         swap elements at spots left and rightI
   swap pivot to spot right
   return location of pivot
```